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Grid-forming wind turbine control based on novel deloading strategy and adaptive configurable natural droop controller

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Weak grids face critical challenges of insufficient inertia support and phase-locked loop instability in wind power integration. Traditional deloading strategies rely on inflexible look-up tables, while virtual synchronous generator (VSG) control suffers from coupled droop and damping parameters, causing severe DC bus voltage fluctuations. To address these issues, this paper proposes a novel deloading strategy coordinating rotor speed and pitch angle via a power conversion coefficient loop, eliminating look-up table dependence. An adaptive configurable natural droop (ACND) controller is further developed to independently adjust inertia, damping and droop coefficients, integrated with a DC voltage loop for power balance. Matlab/Simulink simulations demonstrate that the proposed method achieves stable full-speed deloading, and significantly reduces DC bus voltage fluctuations compared with VSG and matching control, maintaining stability even at low wind speeds with zero reserve power. This control scheme enhances the flexibility and stability of grid-forming wind turbines, providing a reliable solution for high-renewable weak grids.

KEYWORDS
configurable natural droop, deloading control, frequency regulation, gird-forming control, wind turbine

1 Introduction

The growth of renewable energy has outpaced expectations, with wind power emerging as one of the most competitive and fastest-growing technologies due to its low cost and mature technology, making it ideal for large-scale deployment. However, large wind farms are usually situated in remote areas rich in wind resources but distant from the main power grid, leading to weak grid characteristics (Kroposki et al., 2017; Fernández-Guillamón et al., 2019).

Traditional power systems consist of numerous synchronous generators (SGs), which provide inertia by releasing rotor kinetic energy during grid frequency events, while governors regulate primary frequency control (Milano et al., 2018; Matevosyan et al., 2023). Replacing SGs with renewable energy sources connected through power

electronic devices fundamentally alters the operation, control, and stability of power systems.

Most wind turbines (WTs) currently adopt grid-following (GFL) control, where synchronization with the grid is achieved by using a phase-locked loop (PLL) to capture the AC voltage at the point of common coupling (PCC), assuming stable voltage at the PCC. However, with the increasing integration of GFL-interfaced resources, grid stability and reliability are increasingly compromised, and the use of PLL in weak grids leads to oscillation issues (Xu et al., 2019; Hu et al., 2024).

To address these challenges, grid-forming (GFM) control has been envisioned as the cornerstone of future power systems (Rocabert et al., 2012; Tayyebi et al., 2020). Like SGs, GFM control inherently supports power or inertia synchronization, allowing self-synchronization with the grid without additional PLL components and actively supporting grid frequency and voltage. These characteristics make GFM control a promising solution for weak, low-inertia systems, drawing widespread attention as a key technology for future grid integration of wind power and other renewable energy sources (Poolla et al., 2019; Lasseter et al., 2020). Given the increasing prevalence of permanent magnet synchronous generator (PMSG)-based wind turbines, this paper focuses on the analysis and validation of PMSG-based turbines.

For wind turbine to implement grid-forming technology, two key points must be addressed: 1) To simulate the frequency support capability of SGs, WT needs energy regulation capabilities, allowing them to release or absorb power during frequency events. 2) The converter must achieve self-synchronization with the grid without relying on a phase-locked loop, which can be accomplished through GFM algorithms.

Regarding the first point, the deployment of distributed battery energy storage systems and distributed energy resources has proven to be an exceptionally powerful approach, playing an indispensable role in providing critical ancillary services and actively mitigating voltage and frequency deviations (Ma et al., 2017; Mejia-Ruiz et al., 2022). Furthermore, the profound capability of integrated DERs across transmission and distribution networks to deliver highly coordinated frequency and voltage support has been thoroughly validated through real-time co-simulations, showcasing their superior dynamic performance (Mejia-Ruiz et al., 2023). However, while installing additional physical energy storage units or coordinating widespread DERs yields optimal grid support, it inevitably increases the overall capital and communication costs. Therefore, as a more economical and self-contained alternative, utilizing the WT itself to adjust its operating point for power reservation holds significant research value and presents a highly promising solution for future wind power integration. To achieve deloading operation, there are two main options for individual WT: rotor speed control (RSC) (Boyle et al., 2021; Bao et al., 2023) and pitch angle control (PAC) (Singh et al., 2013). RSC achieves deloading by increasing rotor speed but is limited by the maximum speed. PAC reduces the wind energy utilization coefficient by adjusting the pitch angle, but frequent pitch adjustments can cause mechanical wear. Chang-Chien et al. (2011) proposes a strategy that coordinates rotor speed and pitch angle control, using RSC in low wind speeds and PAC in high wind speeds to achieve deloading. However, this strategy does not account for constant-speed regions. Zhang et al. (2018), Zhang et al. (2012) consider the free-running region, constant-speed region, and constant-power region of the WT. In the limited-speed region, there is no direct mapping between rotor speed and active power, making traditional pitch control insufficient for accurate deloading. Thus, traditional methods calculate rotor speed and pitch angle values for various wind speeds and deloading coefficients, constructing look-up tables for coordinated deloading. However, such tables are typically based on specific turbine data and conditions, lacking flexibility to adapt to changing operational environments and increasing memory consumption. However, the deloading control based on a look-up table presents a significant problem of excessive memory consumption. In the conventional deloading control of wind turbines, the core input variables of the look-up table are wind speed and deloading coefficient, and the corresponding output is deloading power. When the value range of deloading coefficient is set to 0.9–0.99 with a step size of 0.01, and the wind speed ranges from 8 m/s to 12 m/s with a step size of 0.0001 m/s, the data storage scale required by the look-up table will reach $10 \times 40{,}000$. This occupies a large amount of hardware storage resources and increases the computational burden of the control program. Kayedpour et al. (2023) explores intelligent control algorithms to achieve WT deloading.

Regarding the second key point, the commonly used GFM control methods for WT include Matching Control (MC) (Huang et al., 2017; Lyu and Groß, 2024) and Virtual Synchronous Generator (VSG) control (Xi et al., 2018; Xu et al., 2021). MC is based on the similarity and duality between the DC capacitor voltage and the SG rotational speed structure and characteristics. By controlling the DC capacitor voltage, it provides system inertia and achieves inverter synchronization with the grid. However, in practical systems, the DC capacitor has limited capacity, offering only a small amount of inertia. Additionally, the nature of MC relies on the DC voltage to track the grid frequency, meaning that any system frequency deviation will cause the DC bus voltage to deviate from its nominal value. VSG control simulates the swing equation of SGs within the inverter, providing similar electromagnetic characteristics, rotor inertia, and both frequency and voltage regulation capabilities (Zhong and Weiss, 2011). However, under low wind speed conditions and during frequency drop events, a fixed-parameter VSG produces a large instantaneous output power, while the turbine's captured power remains small, creating a significant power deficit that leads to large fluctuations in the DC bus voltage. The stability of direct-drive permanent magnet wind turbines, which depend on DC capacitors to connect two inverters, is thus closely tied to the stability of the DC voltage (Tayyebi et al., 2020).

To balance the differences between the turbine's captured power and the VSG's output power, adaptive adjustment of the VSG parameters is necessary. However, since the droop and damping characteristics are coupled, it is difficult to set both to optimal values simultaneously, often requiring a compromise. The turbine's reserve power fluctuates with wind speed and dispatch instructions. When the reserved power is very small or zero, the droop coefficient and inertia constant need to be adjusted to near-zero to balance the captured and output power. However, because the VSG controller has a narrow parameter adjustment range, it may lose stability. Wu et al. (2016), Wen et al. (2025) proposed an

TABLE 1 The wind speed region and the corresponding calculation formuals for C_{pmax} and C_p.

Wind speed region	Calculation formula for C_{pmax}	Calculation formula for C_p
$v_{in} \le v < v^*_{\omega max}$	Equation 14	Equation 3
$v^*_{\omega max} \le v < v_{\omega max}$		
$v_{\omega max} \le v < v_n$	Equation 17	
$v_n \le v$	Equation 18	

adaptive VSG control strategy considering the range of droop coefficients to balance the reserve power and support power of wind turbines.

Zhang et al. (2016) introduces a configurable natural droop (CND) controller, which allows independent configuration of damping, droop coefficient, and inertia constant, simultaneously meeting both dynamic and frequency regulation requirements. Compared to VSG control, CND control offers greater flexibility in adjusting the droop coefficient and inertia constant over a wide range while maintaining a stable damping ratio, making it an ideal grid-forming algorithm for turbines experiencing random power fluctuations.

The comparison between typical existing works and the proposed approach is reported in Table 1.

Inspired by the above research, this paper proposes a grid-forming wind turbine control method based on a novel deloading strategy (NDS) and an adaptive configurable natural droop (ACND) controller. The contributions of this work can be summarized as follows:

1. A novel deloading strategy for WTs is proposed, which considers the free-running region, constant-speed region, and constant-power region. By introducing a wind energy utilization coefficient control loop, rotor speed and pitch angle are coordinated to achieve full-speed deloading of WT without the need for look-up tables or large datasets for model training. This enables the WT to flexibly reserve or release power based on dispatch instructions or grid frequency.
2. A grid-forming control algorithm for deloaded WT is developed based on the CND controller. The inertia constant, damping ratio, and droop characteristics of the CND controller are tuned according to the maximum output power and reserve power, ensuring that the output and input power remain as balanced as possible, thus reducing DC bus voltage fluctuations. Additionally, a DC voltage loop is integrated into the CND control loop, providing compensation to the power outer loop, which helps balance the input and output power, further reducing DC bus voltage fluctuations. During system frequency drops, the capacitor voltage can gradually return to its nominal value without deviation.

The remainder of this paper is as follows. Section 2 introduces the dynamic characteristics of the WT. Section 3 outlines the operational principles and features of CND controller. Section 4 explains the principles of the novel wind turbine deloading strategy. Section 5 discusses the design of the adaptive configurable natural droop controller. Section 6 validates the effectiveness of the proposed strategies. Section 7 provides the conclusion.

2 Operational characteristics of wind turbine

Airflow generates wind energy, and according to Betz's theory, wind turbine blades can capture a portion of the wind power. The power characteristic of a WT is modeled as Equation 1 (Slootweg et al., 2001)

$$P_m = \frac{1}{2}\rho S v^3 C_p(\lambda,\beta) \tag{1}$$

where P_m denotes the turbine mechanical power. ρ is the air density. $S = \pi R^2$ is the effective area swept by the wind turbine blades. R is the blade radius. v is the wind speed. $C_p(\lambda,\beta)$ is the power conversion coefficient. λ is the tip speed ratio, and β is the pitch angle. The definition of the tip speed ratio λ is shown in Equation 2.

$$\lambda = \frac{\omega_r R}{v} \tag{2}$$

ω_r is rotor speed. The characteristics of the wind turbine mainly depend on C_p, which are generally represented by a fitting function (Slootweg et al., 2003). It is important to note that the form of the fitting function is not universal. For different WTs, the fitting function form also varies. The fitting function used in this article is shown in Equation 3.

$$\begin{aligned} C_p(\lambda,\beta) &= 0.645\left\{0.00912\lambda + \frac{116}{\alpha} - 0.4(\beta+2.5) - 5\right\} e^{\frac{-21}{\alpha}} \\ \frac{1}{\alpha} &= \frac{1}{\lambda + 0.08(\beta+2.5)} - \frac{0.035}{(\beta+2.5)^3+1} \end{aligned} \tag{3}$$

$C_p(\lambda,\beta)$ depends on λ and β, and the curve describing their relationship is known as the wind turbine characteristic curve.

As illustrated in Figure 1. With a constant pitch angle β, C_p functions as a single-variable function of λ. It initially increases and then decreases as λ increases, leading to an optimal tip speed ratio λ_{opt} (see Figure 1), where the power conversion coefficient reaches its maximum value C_{pmax}. When the tip speed ratio λ is fixed, it solely relates to the pitch angle β, with C_p diminishing as β increases. As the pitch angle increases, the value of the λ_{opt} decreases accordingly. It should be noted that the maximum power conversion coefficient C_{pmax} used in this paper refers to the maximum value that can be achieved by the wind turbine under constraints such as pitch angle, speed limit value, and power limit value. It does not simply represent the value of C_{pmax} at $\beta = 0°$. Similarly, the maximum

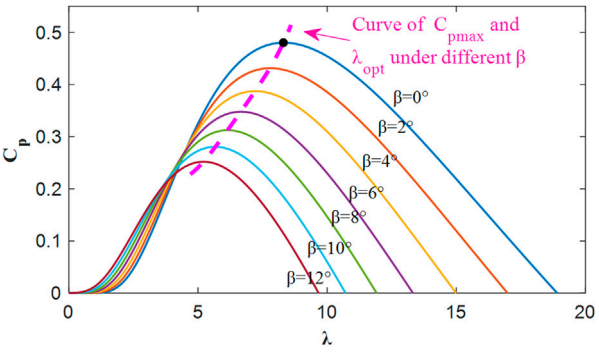


FIGURE 1
Power coefficient Cp as function of the pitch angle β and tip speed ratio λ.

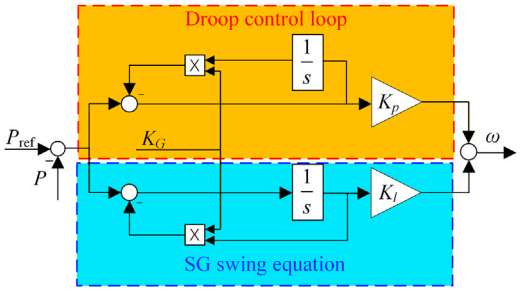


FIGURE 2
Structure of the configurable natural droop controller.

output power P_{mpp} of the wind turbine mentioned below refers to the maximum mechanical power that the wind turbine can output under constrained conditions.

3 Basic principles of the configurable natural droop controller

As illustrated in Figure 2, the structure of the CND controller uses the deviation between the active power reference P_{ref} and the active power P as its input, with the virtual angular velocity ω as its output. The parameters K_P, K_I, and K_G are adjustable. Compared to the conventional VSG control structure, the CND controller offers additional degrees of freedom by allowing independent configuration of the inertia constant, damping characteristics, and droop characteristics, without increasing the order of the transfer function (Zhang et al., 2016). The transfer function of the CND controller is shown in Equation 4.

$$G_{CND}(s) = \frac{\Delta P}{\omega}(s) = \frac{K_P s + K_I}{s + K_G} \tag{4}$$

The block diagram of the active power control loop for the WT's grid-side converter is shown in Figure 3, where θ_{gird} represents the grid phase angle, and δ is the difference between the virtual phase angle θ of the inverter output and the grid phase angle θ_{gird}. P_{max} is a function that represents the relationship between the inverter's power angle and output power. It is worth noting that the voltage and current loops of the converter have a much faster response than the outer power loop, so the effect of the inner loop is neglected here. When the system impedance is approximately inductive, P_{max} can be expressed as Equation 5.

$$P_{max} = \frac{EV}{X} \tag{5}$$

In the equation, E and V represent the RMS values of the inverter output voltage and the grid voltage, respectively, and X denotes the system impedance. By substituting Equation 4 into the transfer function in Figure 3, the closed-loop transfer function for active power is obtained as shown in Equation 6, and the P-f characteristics are shown in Equation 7, where ξ represents the damping characteristic, and ω_n is the natural frequency.

$$\frac{P}{P_{ref}}(s) = \frac{(2\xi\omega_n - K_G)s + \omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} \tag{6}$$

$$\frac{\Delta P}{\Delta\omega_g}(s) = \frac{-P_{max}(s + K_G)}{s^2 + 2\xi\omega_n s + \omega_n^2} \tag{7}$$

The relationship between the inertia constant H of a conventional SG and the natural frequency ω_n can be shown as Equation 8 (Zhang et al., 2016)

$$H = \frac{\omega_s}{2K_I S_N} \tag{8}$$

ω_s represents the synchronous angular frequency, and S_n denotes the rated capacity of the inverter. From Equations 6–8, the relationships between K_I, K_G, K_P, and the system's damping coefficient, inertia constant, and droop coefficient D_P are obtained as follows:

$$K_I = \frac{\omega_s}{2HS_N} \tag{9}$$

$$K_G = D_p K_I \tag{10}$$

$$K_P = \frac{2\xi\sqrt{P_{max}K_I} - K_G}{P_{max}} \tag{11}$$

First, the values of H, ξ, and D_P are determined based on the actual system conditions. Then, the values of K_I, K_G, and K_P are calculated using Equations 9–11 and input into the CND controller. This allows for the independent configuration of the inertia constant, damping characteristics, and droop coefficient.

4 Novel deloading strategy for wind turbines

4.1 Operating principle of the novel deloading strategy

As shown in Figure 4, the NDS coordinates the turbine speed and pitch angle through the power conversion coefficient C_p control loop. The switching between RSC mode and PAC mode is determined by the trigger condition, where RSC mode is

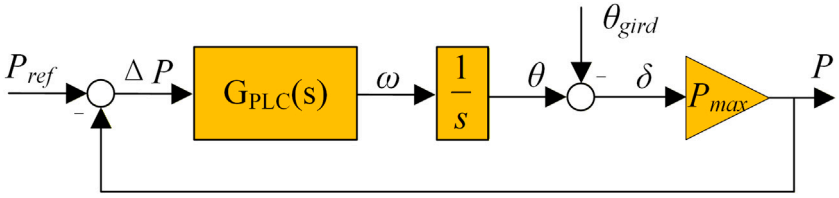


FIGURE 3
Active power loop block diagram of the configurable natural droop controller.

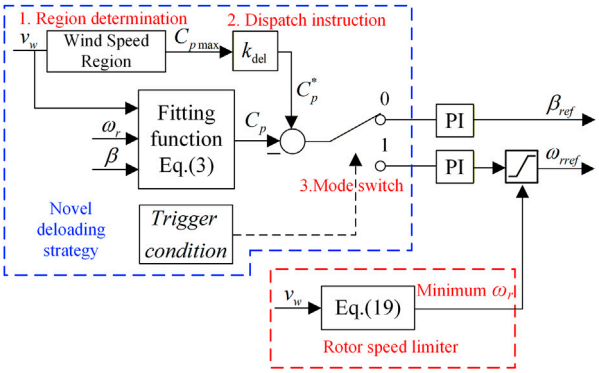


FIGURE 4
Control block diagram of the novel deloading strategy for WT.

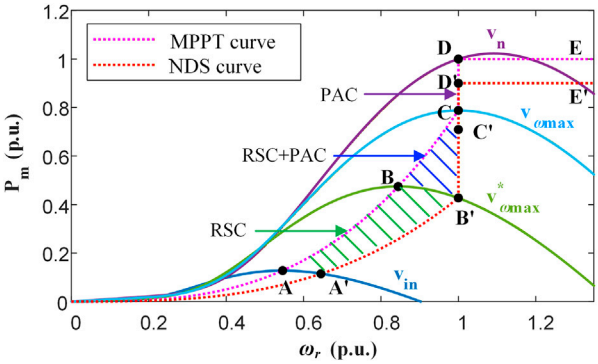


FIGURE 5
Operating characteristics of WT under MPPT mode and NDS mode across full wind speed range.

represented by 1 and PAC mode by 0. The specific trigger condition is given by (Equation 12).

$$Trigger\ condition = \begin{cases} 0 & \text{mode} = 1 \&\& \omega_{rref} \geq \omega_{r\,\max} \&\& C_p > C_p^* \\ 1 & \text{mode} = 0 \&\& \beta \leq 0 \&\& C_p < C_p^* \end{cases} \tag{12}$$

In this work, we define the deloading coefficient k_{del}. The reserve power of the wind turbine P_{del}=(1-k_{del})P_{mpp}. The smaller the k_{del}, the more power is relatively reserved by the WT. Additionally, by changing the value of k_{del}, the release and reservation of WT reserve power is achieved.

To fully utilize the rotor's kinetic energy, the NDS prioritizes deloading by increasing the rotor speed. When the rotor speed reaches its upper limit, the pitch angle control is activated for further deloading. Figure 5 shows a comparison between the MPPT operation and the deloading control of a variable-speed WT under full wind speed conditions.

As shown in Equation 1, it can be seen that in the WT's mechanical power equation, $1/2\rho S v_w^3$ is an uncontrollable parameter that depends on the turbine's mechanical structure and wind speed. Therefore, the essence of NDS is to control the value of $C_p(\lambda,\beta)$ to track the command C_p^* by changing λ and β. Define deloading power conversion coefficient C_{pdel} see Equation 13.

$$C_{p\,\mathrm{del}}(\lambda_{\mathrm{del}},\beta_{\mathrm{del}}) = k_{del} C_{p\,\max} \tag{13}$$

During the deloading process, the operational range of the rotor speed and pitch angle is determined by wind speed. The deloading characteristics of the WT vary in different wind speed regions (Tu et al., 2022).

4.1.1 Low wind speed region ($v_{in} \leq v < v^*_{\omega max}$)

When the wind turbine operates in MPPT mode (see segment AB in Figure 5), the tip speed ratio tracks the optimal value λ_{opt}, which is generally given by the WT manufacturer. At this time, the maximum value of C_p is shown in Equation 14.

$$C_{p\,\max} = C_p(\lambda_{\mathrm{opt}},0) \tag{14}$$

When the wind speed is low, the operating speed of the WT is also low, providing sufficient speed margin for deloading through overspeed operation. At this point, mode switch is set to 1, and the NDS achieves active power reservation and release solely through RSC by increasing the tip-speed ratio to λ_{del}, with no pitch angle adjustment, resulting in a change in the power , as shown in Equation 15.

$$C_{p\,\mathrm{del}} = C_p(\lambda_{\mathrm{del}},0) \tag{15}$$

λ_{del} represents the tip-speed ratio that must be satisfied when the pitch angle is zero and $C_p = C_{pdel}$.

Due to deloading operation using overspeed methods, WT will reach its rated speed at lower wind speeds (see segment A′B′ in Figure 5), this wind speed is denoted as $v^*_{\omega max}$. The value of $v^*_{\omega max}$ is related to the set deloading coefficient k_{del}. For example, the smaller the k_{del}, the smaller the corresponding minimum value

of C_{pdel}, and the larger the tip-speed ratio required for deloading, resulting in a greater increase in turbine speed. As the wind speed increases, the rotor speed reaches the limit value more quickly, thus reducing the value of $v^*_{\omega max}$.

4.1.2 Medium wind speed region ($v^*_{\omega max} \le v < v_{\omega max}$)

In the medium wind speed region, when the WT operates in MPPT mode and the rotor speed has not reached its limit (see segment BC in Figure 5), the C_{pmax} is calculated in the same manner as Equation 14.

When the wind speed reaches $v^*_{\omega max}$, the rotor speed of the wind turbine with the deloading coefficient k_{del} has already reached its upper limit. Under the NDS mode, the WT activates PAC to continue deloading (see segment B′C′ of Figure 5). The deloading power conversion coefficient is

$$C_{pdel} = C_p\left(\frac{\omega_{r\,max}R}{v}, \beta_{del}\right) \tag{16}$$

β_{del} represents the pitch angle required to satisfy $C_p = C_{pdel}$ when the wind turbine is operating at maximum rotor speed.

In the medium wind speed range, the rotor speed does not reach the upper limit in MPPT mode. Therefore, to reserve and release active power, the WT requires coordination between RSC and PAC. Suppose the WT, initially operating in MPPT mode, receives a deloading command. NDS will prioritize deloading through RSC, and the switch is set to 1. If the rotor speed reaches the upper limit ω_{rmax}, but C_p still does not meet the required deloading coefficient C_p^*, the trigger condition is satisfied, and the switch is set to 0, activating PAC for deloading. When the WT receives a dispatch instruction to release reserved power, NDS will first reduce the pitch angle through PAC. If the pitch angle $\beta = 0$ is detected and C_p is still lower than the target C_p^*, the mode switch is set from 0 to 1, and RSC is used to release the remaining reserved power.

It should be noted that the introduction of $v^*_{\omega max}$ in this paper is to facilitate the explanation of the wind turbine's deloading operation characteristics. The NDS algorithm does not require the calculation or detection of the medium-low wind speed boundary $v^*_{\omega max}$; the transition between RSC and PAC below high wind speeds can be achieved through Equation 12.

4.1.3 High wind speed region ($v_{\omega max} \le v < v_n$)

The characteristic of this region is that when the wind turbine operates in MPPT mode, the speed reaches the upper limit ω_{rmax}, and C_{pmax} is no longer obtained from the optimal tip-speed ratio but gradually becomes less than the optimal tip-speed ratio. Therefore, as the wind speed increases, C_{pmax} slowly decreases, but the WT's mechanical power continues to gradually increase until it reaches the rated power (see segment CD in Figure 5). The expression for the maximum power conversion coefficient is

$$C_{p\,max} = C_p\left(\frac{\omega_{r\,max}R}{v}, 0\right) \tag{17}$$

When the wind turbine is operating in deloading mode and its rotor speed has reached the upper limit, NDS limits the speed to ω_{rmax}, and power reservation and release are carried out solely through PAC (see segment C′D′ of Figure 5), with the mode switch set to 0. The deloading power conversion coefficient is the same as in Equation 16.

4.1.4 Power-limited wind speed region ($v_n \le v$)

When $v_n \le v$, the maximum wind energy captured by the turbine exceeds its rated power, leading to the activation of the pitch angle to limit power capture (see segment DE in Figure 5). At this moment, the maximum value of the power conversion coefficient is

$$C_{p\,max} = \frac{2P_n}{\rho\pi R^2 v^3} \tag{18}$$

At this stage, NDS continues to reserve and release power through PAC (see segment D′E′ of Figure 5), and the deloading power conversion coefficient remains the same as in Equation 16.

4.2 Operating method of the novel deloading strategy

First, set the deloading coefficient k_{del} according to the station instructions and two wind speed thresholds: $v_{\omega max}$ and v_n. Subsequently, based on Equations 14, 17, 18, the maximum value of the power conversion coefficient C_{pmax} is derived. The reference value of power conversion coefficient C_p^* is determined by setting the deloading command k_{del} and C_{pmax}. Based on Equation 3, the current power conversion coefficient C_p is calculated. Finally, based on the Equation 12, the difference between the C_p^* and C_p is input into the RSC or PAC, allowing the C_p to track the target C_p^*, thereby enabling the wind turbine to perform deloading operation according to the command. The wind speed region and the corresponding calculation formulas for C_{pmax} and C_p are shown in Table 2. This strategy eliminates the need to construct a look-up table, thereby avoiding the data memory occupation with a scale of 10 × 40,000 required by the conventional deloading control.

4.3 Minimum speed limit of NDS

Since there is a risk of small-signal instability when the WT operates on the left side of the MPPT point, most turbines are operated on the right side of the MPPT point (Arani and Mohamed, 2015). NDS defaults to adjusting the turbine power on the right half of the dynamic curve (i.e., $\omega_r > \omega_{opt}$). Thus, when $C_p < C_p^*$, NDS reduces the turbine speed, and conversely, when $C_p > C_p^*$, NDS increases the speed. However, the relationship between C_p and turbine speed ω_r is not a one-to-one mapping. As shown in Figure 6, within the instability speed region (i.e., ω_r < Instability speed), when $C_p < C_p^*$, the turbine speed controlled by NDS continues to decrease, leading to uncontrolled speed reduction due to positive feedback. Therefore, the minimum turbine speed reference must be constrained. For conservative reasons, the lower limit of turbine speed is set to ω_{opt} in this paper. The value of ω_{opt} varies depending on the wind speed, and the specific values are given by Equation 19.

$$\begin{cases} \omega_{rref} < \omega_{opt} & v \le v_{\omega\,max} \\ \omega_{rref} < \omega_{r\,max} & v > v_{\omega\,max} \end{cases} \tag{19}$$

TABLE 2 Comparison of between the exisiting and proposed methods.

No.	Source of methods	Rotor speed limitations	Consider mechanical wears	No look-up table required	Variable parameter	DC bus voltage stability
1	Singh et al. (2013)	—	N	N	—	—
2	Chang-Chien et al. (2011)	N	Y	N	—	—
3	Zhang et al. (2016)	Y	Y	N	—	—
4	Lyu and Groß (2024)	Y	Y	N	—	N
5	Xi et al. (2018)	—	—	—	N	N
6	Proposed	Y	Y	Y	Y	Y

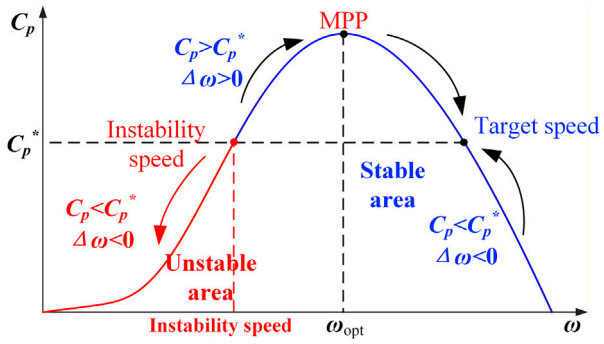


FIGURE 6
Rotor speed instability in NDS.

5 Design of the adaptive CND controller

Given the uncertainty in wind turbine output power, during a frequency drop event, the imbalance between the output power of the grid-side converter controlled by a fixed-parameter controller and the WT's captured power can lead to DC bus voltage fluctuations, which, in severe cases, may result in turbine system collapse. Given the uncertainty in WT output power, during a frequency drop event, the imbalance between the output power of the grid-side converter controlled by a fixed-parameter controller and the WT's captured power can lead to DC bus voltage fluctuations, which, in severe cases, may result in turbine system collapse. To mitigate DC bus voltage fluctuations, an adaptive parameter design for the CND controller is necessary to ensure that the output power of the grid-side converter aligns as closely as possible with the wind turbine's output power.

5.1 Adaptive control of the droop coefficient

During a frequency drop event, the WT releases a certain proportion of its reserve power based on the frequency deviation to support the grid. To ensure that the droop power output of the CND controller aligns with that of the wind turbine, its droop coefficient must be equivalent to the wind turbine's effective droop coefficient.

This paper uses the NDS algorithm to calculate the current reserve power P_{del} of the wind turbine in real time, and the droop coefficient D_P is adaptively adjusted based on this value.

$$D_a = \frac{P_{del}}{\Delta\omega} \tag{20}$$

$\Delta\omega$ represents the deviation between the grid frequency and the rated frequency, with its value set to $2\pi\times0.5$. This equation indicates that the wind turbine releases the full reserve power when the grid frequency drops by 0.5 Hz.

5.2 Adaptive control of the inertia constant

The inertia constant H reflects the generator's ability to resist frequency changes by relying on its rotational inertia to provide short-term mechanical energy during system disturbances. The inertia constant is related to energy; the larger the inertia constant, the more energy the generator releases. Since the WT's captured power fluctuates with wind speed, it can be considered as a generator with momentarily changing rated power. Thus, the rated power of the wind turbine cannot be used to set the inertia constant. For the CND controller, the inertia constant formula is shown in Equation 8. Based on the maximum output power P_{mpp} calculated by the NDS algorithm, this paper treats it as the equivalent rated power of the WT at that moment and adaptively controls the inertia constant H. By substituting P_{mpp} into Equation 8, the following result is obtained.

$$H_a = \frac{\omega_s}{2K_I P_{mpp}} \tag{21}$$

5.3 Small-signal analysis

Equation 5 is expanded by small-signal perturbation at the steady-state operating point (δ_0, P_0), as shown in Equation 22:

$$\Delta P_0 + \Delta P = \frac{EV}{X}\sin(\delta_0 + \Delta\delta) \approx \frac{EV}{X}(\sin\delta_0 + \cos\delta_0 \cdot \Delta\delta) \tag{22}$$

The incremental components are extracted to obtain the linearized output power deviation, as shown in Equation 23:

$$\Delta P = K_S \Delta\delta \tag{23}$$

where $\Delta\delta$ denotes the power angle deviation between the inverter and the power grid; Ks is the synchronous power coefficient, and $K\text{s} = EV\cos\delta_0/X$. A weaker grid strength (i.e., a lower Short-Circuit Ratio (SCR) and a larger equivalent impedance X) results in a smaller value of Ks. Combined with the control strategy proposed in this paper, the active power deviation is substituted into the closed-loop electromechanical swing equation, and the dynamic equation of virtual angular frequency with adaptive parameters can be derived as Equation 24:

$$2H_a\frac{d(\Delta\omega)}{dt} = \Delta P_{ref} - K_S\Delta\delta - D_{active}\Delta\omega \tag{24}$$

where D_{active} represents the effective active damping of the system. The state variable vector is selected as $x = [\Delta\delta,\Delta\omega]^T$, and the core state matrix A of the closed-loop system can be deduced as Equation 25:

$$A = \begin{bmatrix} 0 & 1 \\ -\frac{K_S}{2H_a} & -\frac{D_{active}}{2H_a} \end{bmatrix} \tag{25}$$

The closed-loop small-signal stability and dynamic response characteristics of the system are determined by the characteristic equation of matrix A, which is expressed as Equation 26:

$$s^2 + \frac{D_{active}}{2H_a}s + \frac{K_S}{2H_a} = 0 \tag{26}$$

According to the eigenvalue sensitivity analysis, in the conventional VSG control, the effective damping of the system is coupled with the droop coefficient, and the distribution of its dominant eigenvalues λ_{VSG} is given by:

$$\lambda_{VSG} = -\frac{D_V}{4H_V} \pm \frac{1}{2}\sqrt{\left(\frac{D_V}{2H_V}\right)^2 - \frac{2K_S}{H_V}} \tag{27}$$

where H_V and D_V are the virtual inertia and droop coefficient of the VSG control, respectively. The instability mechanism of the conventional VSG can be intuitively observed from Equation 27: when the reserve power of the wind turbine is completely exhausted, forcing the droop coefficient D_P to approach 0, the real parts of the poles will rapidly tend to zero. As shown by the root locus in Figure 7b, the poles of the conventional VSG drift drastically toward the imaginary axis with the decrease of D_P, which indicates that the system is prone to losing the stability margin and inducing severe oscillations under the condition of zero reserve power.

In contrast, the ACND controller proposed in this paper completely overcomes this defect through a parameter decoupling mechanism. To ensure that the system can maintain the optimal dynamic damping ratio ξ (e.g., $\xi = 0.707$) under all operating conditions, the equivalent active damping actually provided by the ACND is strictly and adaptively matched as:

$$D_{active} = 2\xi\sqrt{2H_aK_S} \tag{28}$$

By substituting Equation 28 into the characteristic equation for solution, the dominant eigenvalues λ_{ACND} of the system under ACND control can be obtained as:

$$\lambda_{ACND} = -\xi\sqrt{\frac{K_S}{2H_a}} \pm j\sqrt{1-\xi^2}\sqrt{\frac{K_S}{2H_a}} \tag{29}$$

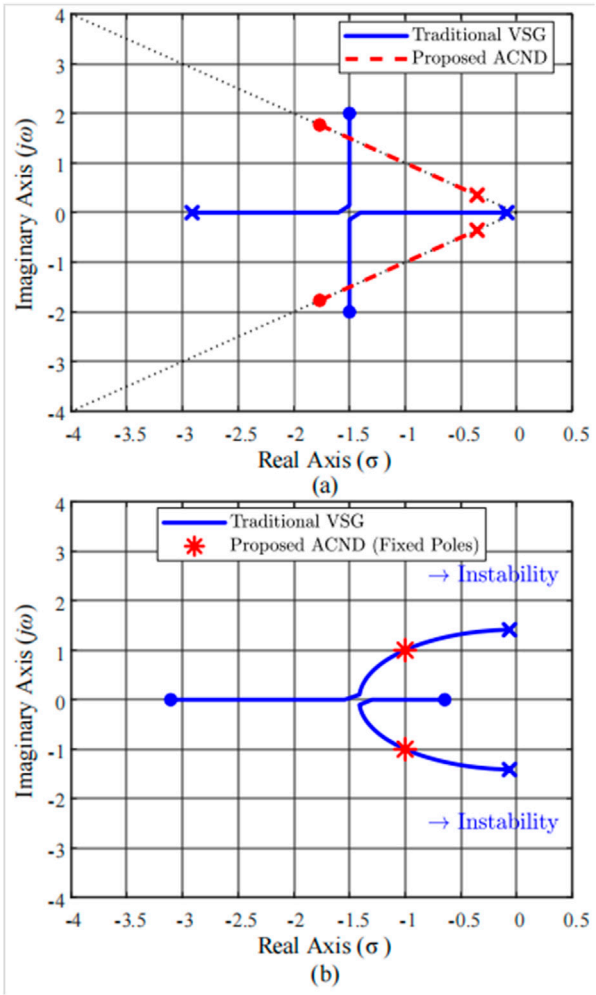


FIGURE 7
Eigenvalue analysis. **(a)** Trajectory with decreasing grid strength. **(b)** Trajectory with decreasing reserve power.

Equation 29 provides a rigorous mathematical proof for the system stability. Through the comparison between Figures 7a,b, it can be found that since the damping attenuation term of the ACND is forcibly anchored to the target damping ratio ξ, the dominant poles of the ACND are always locked on the preset constant-damping ray or fixed in the safe region of the left half-plane, regardless of the fluctuation of grid strength or the drastic adjustment of the steady-state droop coefficient of the wind turbine (even when the reserve power is reduced to zero).

Even in the extreme parameter-sensitive region with an extremely weak grid, the ACND can still ensure that the eigenvalues are stably retained in the left half of the complex plane by virtue of the decoupled damping control dimension, thus exhibiting excellent robustness.

5.4 DC voltage control loop

The reference power of the CND controller is set to the mechanical power P_m generated by the WT (see Figure 8), which

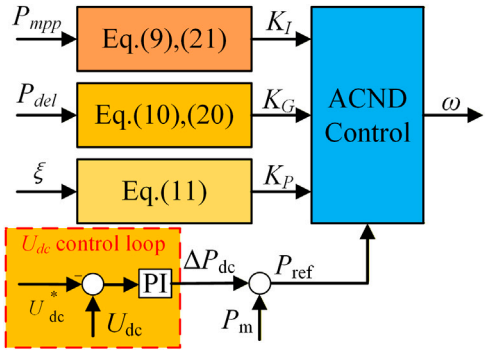


FIGURE 8
Control block diagram of the adaptive CND controller.

can be calculated either through Equation 1 or by measuring the generator's speed and shaft torque. Additionally, a U_{dc} control loop is added to the reference power P_{ref}, creating a coupling between the converter's output power P and the DC bus voltage U_{dc}, and thereby coupling with the wind turbine's output power P_m. Due to the nature of the pitch mechanism, the power release speed of the PAC is slower than that of the RSC, and both the magnitude and speed of the wind turbine's power release are uncertain. As a result, imbalances between the grid-side converter's output power and the turbine's input power can easily occur. The CND adaptive parameter control method can largely match the output and input power, but due to the inertia of both the CND controller and the wind turbine, perfect matching cannot be achieved. When the output and input powers are unequal, the U_{dc} control loop compensates the power reference, ensuring that the output power matches the input power and maintaining the DC bus voltage at its nominal value.

It is worth noting that there are two ways to introduce the U_{dc} control loop in a grid-forming control scheme. One is similar to Matching Control, where the U_{dc} loop is added to the virtual phase angle output side. Alternatively, placing the U_{dc} control loop on the active power reference side provides the VSG controller with stronger small-signal stability under weak grid conditions (Shen et al., 2024).

Based on all the discussions presented above, the advantages of the proposed control strategy over other conventional control strategies are summarized in Table 2.

6 Simulation verification

A typical equivalent system topology shown in Figure 9 is employed in this article and Matlab/Simulink is utilized to validate the performance of the proposed control scheme. The parameters of the wind turbine, bus capacitor, and grid-side filter are listed in Table 3. The upper speed limit ω_{rmax} is set to the rated speed, and the power upper limit is set to the rated power. The wind speed thresholds $v_{\omega max}$ and v_n are set to 11 m/s and 12 m/s, respectively.

6.1 Validation of the effectiveness of the novel deloading strategy

Figure 10 shows the changes in wind speed during this experiment. The full wind speed deloading simulation results for a single 2 MW wind turbine are presented in Figure 11. Key parameters of the deloading control algorithm are displayed, including the rotor speed ω_r, pitch angle β, power conversion coefficient C_p, and the active power output P_m of the wind turbine, compared to the traditional MPPT control strategy. The deloading coefficient k_{del} is set to 0.9. In the low wind speed region (0-t_1), the system achieves the required deloading solely through RSC (see Figure 11b). As the wind speed gradually increases into the medium wind speed region (t_1-t_2), the rotor speed limit necessitates an increase in pitch angle to meet the deloading requirement, causing NDS to switch from RSC to PAC (see Figure 11b). When the wind speed enters the high wind speed region (t_2–t_3), the WT under MPPT control also reaches the rotor speed limit. Due to the speed limit and the increasing wind speed, the tip-speed ratio decreases $\lambda<\lambda_{opt}$, meaning that the power conversion coefficient does not reach the theoretical maximum but is calculated using Equation 17 (see Figure 11c). After $t > t_3$, when the wind speed exceeds 12 m/s, the WT enters the power limitation region, and both MPPT and NDS limit power output using PAC. When the wind speed decreases from 13 m/s to 8 m/s, the dynamic behavior of the wind turbine remains consistent with earlier patterns. It can be seen that under NDS, the power conversion coefficient C_{pdel} maintains a constant 0.9 ratio to the maximum power conversion coefficient C_{pmax}, and the turbine's output power P_m remains 0.9 times the maximum output power P_{mpp}, confirming the effectiveness of the deloading strategy.

The wind turbine reserve power dispatch experiment is shown in Figure 12. At the initial moment, the NDS deloading coefficient is set to k_{del} = 0.9. At $t = t_1$, WT receives a dispatch command to release 60% of the reserve power, and NDS increases the deloading coefficient to k_{del} = 0.96. At this point, the WT is already in the PAC stage, so the pitch angle decreases to respond to the dispatch command (see Figure 12b). At $t = t_2$, the WT receives a dispatch command to release 100% of the reserve power, and the pitch angle further decreases to zero. Since the C_p value does not meet the required C_p^*, PAC switches to RSC, and the turbine speed decreases to release the reserve power. As seen in the figure, after $t = t_3$, the NDS operating characteristics are consistent with the MPPT operating characteristics of the wind turbine. Figures 12c,d show the changes in the power conversion coefficient C_p and the turbine's mechanical power P_m. This experiment verifies that NDS can flexibly release reserve power according to the dispatch command.

To verify the effectiveness of NDS under actual wind speed deloading conditions, this experiment simulates a wide range of randomly fluctuating wind speeds, with the maximum wind speed reaching 13 m/s and the minimum 7 m/s, as shown in Figure 13a. The deloading ratio is set to k_{del} = 0.9. Figures 13b,c show the changes in rotor speed and pitch angle. It can be seen that NDS enables the turbine to maintain a constant deloading ratio between its output power of the wind turbine under randomly fluctuating wind speeds (see Figure 13d), confirming the effectiveness of the proposed deloading strategy.

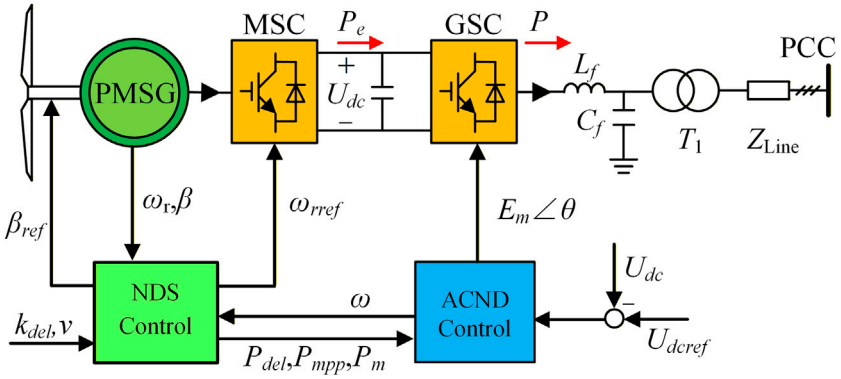


FIGURE 9
Grid-forming wind turbine control structure based on NDS and ACND.

TABLE 3 Parameter for WT system.

Component	Parameter	Value
WT	Rated power	2 MW
	Rated wind speed	12 m/s
	Rated rotor speed	2.463 rad/s
	Inertia time constant	3 s
Converter	DC-link voltage	1,800 V
	DC-link capacitor	100 mF
LCL filter	GSC filter resistance	5 mΩ
	GSC filter inductance	0.317 mH
	GSC filter capacitance	500 μF

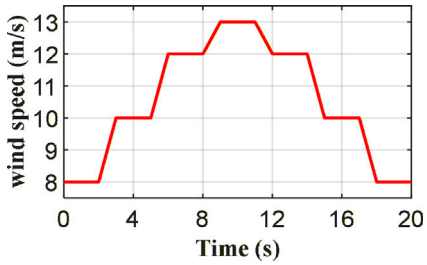


FIGURE 10
Variation of system wind speed.

6.2 Comparison of different grid-forming control algorithms

Wind turbines operating in deloading mode are typically suitable for grids in remote areas with poor power absorption capacity. To further verify the reliability of the proposed strategy, the RT-LAB hardware-in-the-loop simulation platform is adopted in the experiments. The HIL tests use the OP4512 to simulate the IEEE three-machine nine-bus test system, and the control algorithm is implemented based on DSP-TMS320F28335. The output waveforms are collected and displayed by an oscilloscope. As shown in Figure 14, the WT section consists of a 2 MW direct-drive wind turbine. The parameters of the wind turbine are listed in Table 3. G1 and G2 are thermal generators with rated capacities of 4 MW and 2 MW, respectively, and an inertia time constant H_{SG} = 6 s. The wind power penetration is 25%. L1, L2, and L3 are system loads, with values of 2 MW, 1 MW, and 2.5 MW, respectively. At t = 1 s, a 0.75 MW load L4 is suddenly added at bus 4. G1 and G2 regulate the frequency through their governors with a 5% droop coefficient. The WT control method is consistent with that shown in Figure 9, with an initial deloading coefficient k_{del} set to 0.9. The NDS directly receives the virtual angular frequency ω generated by the grid-forming controller to control the release of reserve power, without relying on a PLL to detect the grid frequency. This is one of the key advantages of the grid-forming algorithm. The droop coefficient of the wind turbine is set to $D_P = P_{del}/\pi$, meaning that when the frequency drops by 0.5 Hz. To verify the superiority of the proposed strategy, conventional matching control and adaptive VSG control are adopted for comparison in this experiment, with their detailed structures referred to in Huang et al. (2017) and (Xi et al., 2018). (Zhang et al., 2021) points out that when the inertia constant of the grid-forming controller is too large and the energy at the source end is insufficient, oscillation problems may arise, and the frequency support effect may be inferior to that of grid-forming control without inertia (i.e., Droop Control). Since the energy of the WT is limited by wind speed, the inertia time constants H of the ACND controller and the adaptive VSG controller are both set to 0.2 s. To achieve a better damping ratio for the VSG, its droop coefficient D is set to 500. The ACND controller's damping ratio is set to 0.707, and its droop coefficient is kept equal to that of the wind turbine.

When the wind speed is 12 m/s and load disturbance occurs in the system, the output power of the ACND controller is almost consistent with the wind turbine power because its parameters are adaptively tuned in full accordance with the output characteristics of

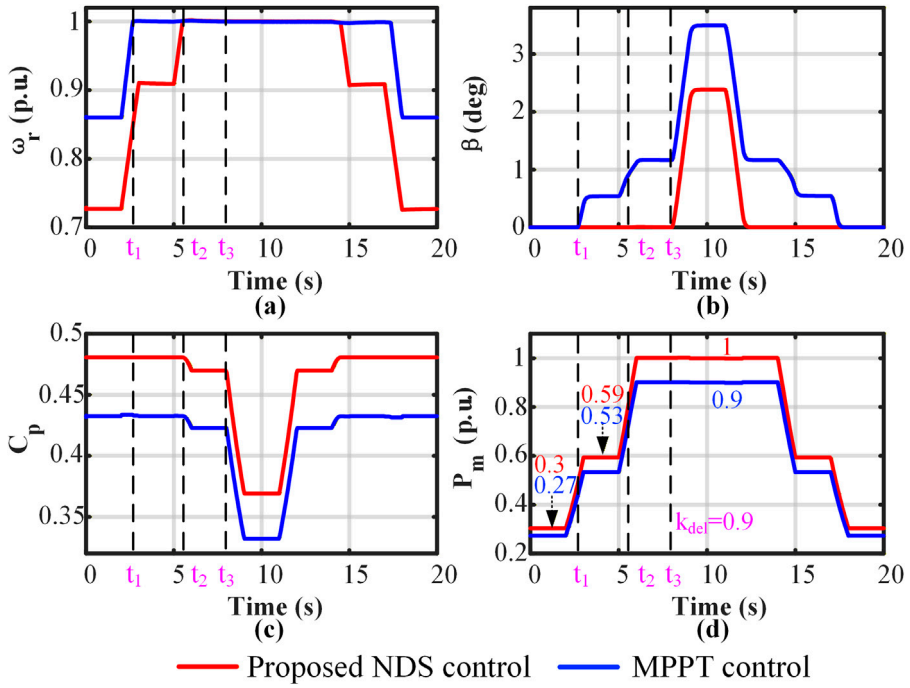


FIGURE 11
Comparison of NDS operation and MPPT Operation with kdel = 0.9 of a single 2 MW WT. **(a)** Rotor speed. **(b)** Pitch angle. **(c)** Power conversion coefficient. **(d)** Mechanical power of WT.

the wind turbine (see Figures 15a,b). The minimum DC bus voltage fluctuation is approximately 1786 V, which gradually recovers to the rated value (see Figure 15c). Since the reserve power is relatively large, the parameter tuning of adaptive VSG is similar to that of ACND under this condition, resulting in similar output power and DC voltage fluctuation characteristics. The output characteristic of matching control is tracking, that is, its output power varies by tracking the input power of the wind turbine. However, the matching controller uses DC voltage to track the system frequency variation, so voltage deviation arises due to frequency deviation. When the system frequency is stable, there is a difference of about 12.5 V between the DC voltage and its rated value. It should be noted that the WT's ability to support grid frequency depends almost entirely on the energy released from the WT, and the influence of the grid-forming control method is minimal. The focus of this paper is not on the WT's frequency support but on better balancing the energy between the WT and the grid-forming control to reduce DC bus voltage fluctuations and enable the WT to stably support the grid frequency.

At a wind speed of 10 m/s, the overall dynamic performance of the wind turbine generator is almost consistent with that at 12 m/s. The difference lies in that the frequency support capability of the wind turbine generator to the system is reduced due to the decrease in its output power and reserve power (see Figures 16a,d). As the reserve power of the wind turbine decreases, the parameter regulation capability of the adaptive VSG approaches its limit. Affected by the feedback regulation of the DC voltage loop, the subsequent output power under the adaptive VSG control presents oscillatory fluctuations (see Figure 16b), which further aggravates the DC bus voltage drop to 1768 V (see Figure 16c). For the matching control, the output power follows the wind turbine power, and the DC voltage tracks the system frequency with a deviation of approximately 12.5 V. The output power P of the proposed ACND controller can accurately track the electromagnetic power P_e of the wind turbine, featuring the minimum DC bus voltage fluctuation without steady-state deviation, thus achieving the optimal comprehensive performance.

At a wind speed of 8 m/s, the output power of the wind turbine decreases further. At this time, the wind turbine operates in the RSC mode, which can instantaneously release a large amount of rotor kinetic energy to provide frequency support.

As the reserve power of the wind turbine decreases, the control parameters of the adaptive VSG reach their limits and cannot be further reduced. After a sudden load increase, the output power fluctuates drastically, causing the DC bus voltage to drop rapidly to 1,750 V. The interaction between the DC voltage loop and the VSG control loop induces large-amplitude power oscillations, which weakens the frequency support capability and results in a system frequency drop to 49.86 Hz (see Figures 17b–d).

The output characteristics of the matching control and the ACND control are almost identical, but the DC voltage still varies with the system frequency. As the system frequency decreases, the DC voltage stabilizes at approximately 1775 V. The output power of the ACND controller can still accurately track the electromagnetic power output by the wind turbine. Since the NDS releases a large amount of rotor kinetic energy for frequency

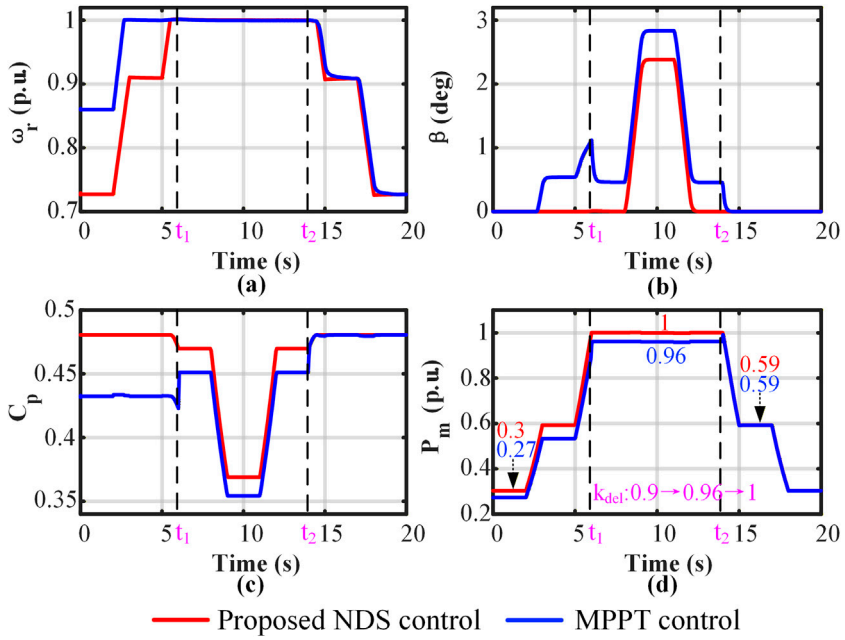


FIGURE 12
NDS experiment of reserve power release according to dispatch instructions. **(a)** Rotor speed. **(b)** Pitch angle. **(c)** Power conversion coefficient. **(d)** Mechanical power of WT.

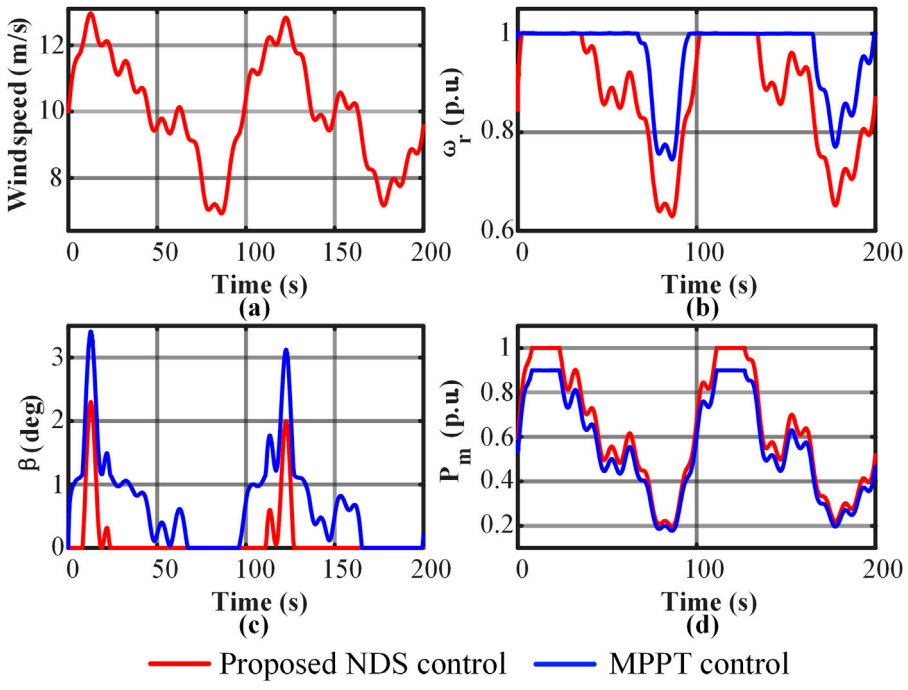


FIGURE 13
NDS deloading operation experiment under simulated real wind speed. **(a)** Real wind speed. **(b)** Rotor speed. **(c)** Pitch angle. **(d)** Mechanical power of WT.

support at this wind speed, the DC voltage of the ACND rises briefly and finally tends to be stable, with its fluctuation amplitude remaining the smallest among the three grid-forming control strategies.

This experiment effectively verifies the superiority of the proposed ACND controller over the VSG controller and the matching controller in terms of power balance performance.

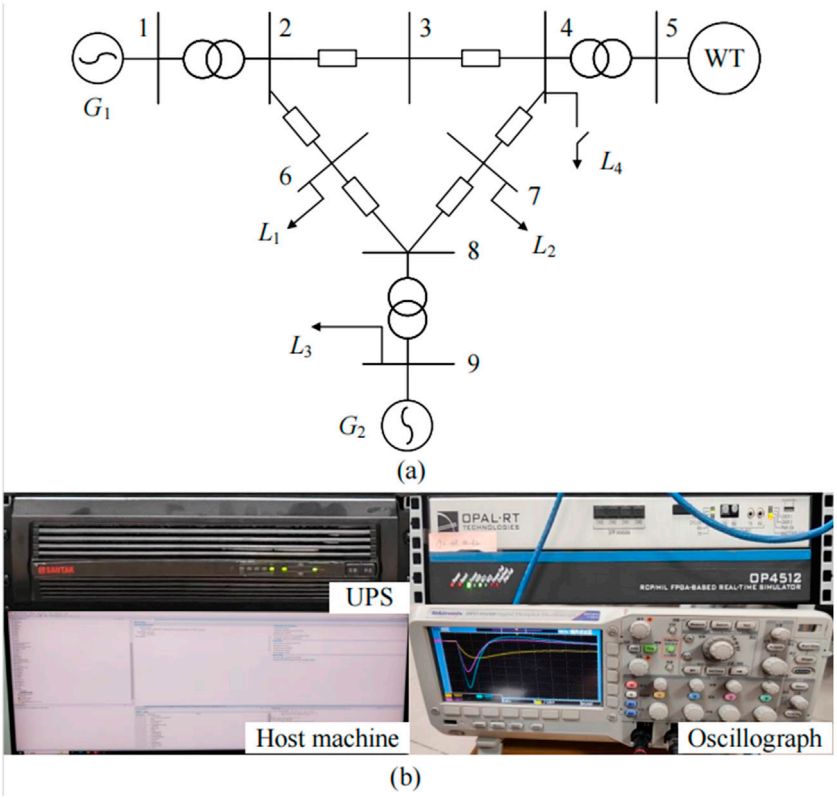


FIGURE 14
Experimental Model and System **(a)** IEEE three-machine nine-bus test model. **(b)** Hardware-in-the-Loop Test System.

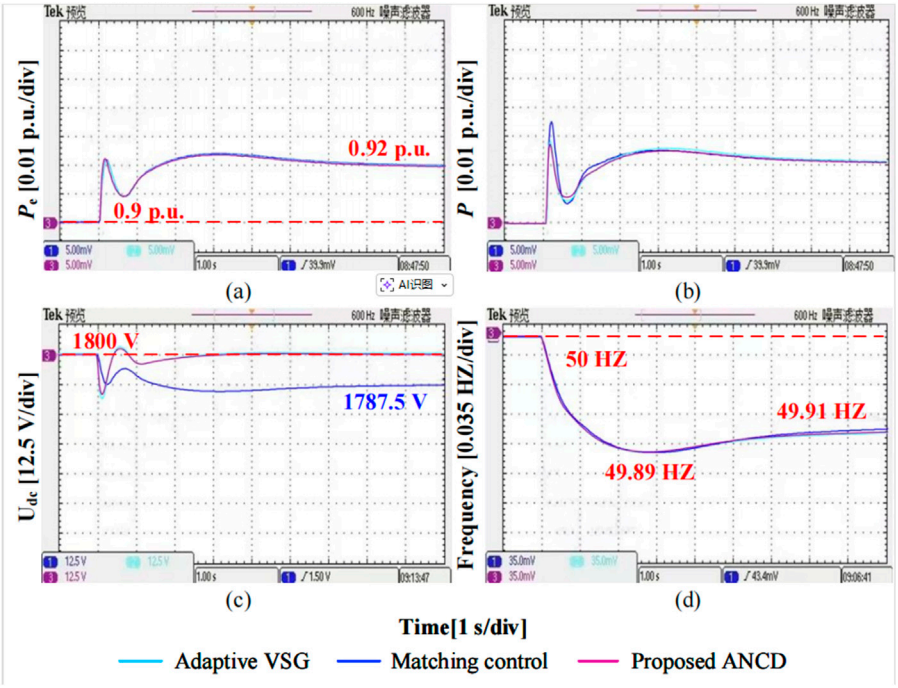


FIGURE 15
Simulation results of different grid-forming control strategies at a wind speed of 12 m/s. **(a)** WT electromagnetic power. **(b)** Grid-side converter power. **(c)** DC bus voltage. **(d)** System frequency.

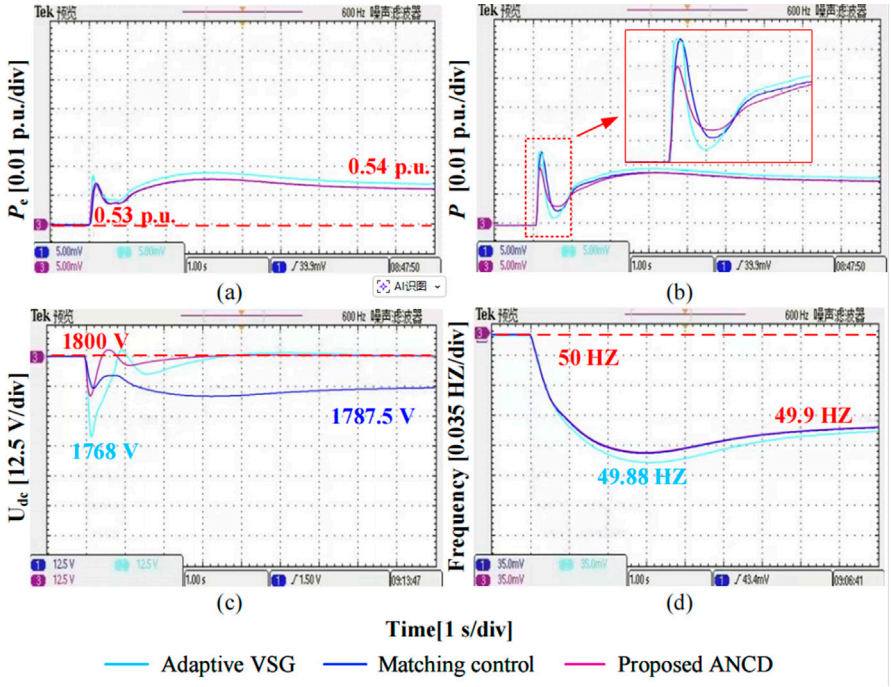


FIGURE 16
Simulation results of different grid-forming control strategies at a wind speed of 10 m/s. **(a)** WT electromagnetic power. **(b)** Grid-side converter power. **(c)** DC bus voltage. **(d)** System frequency.

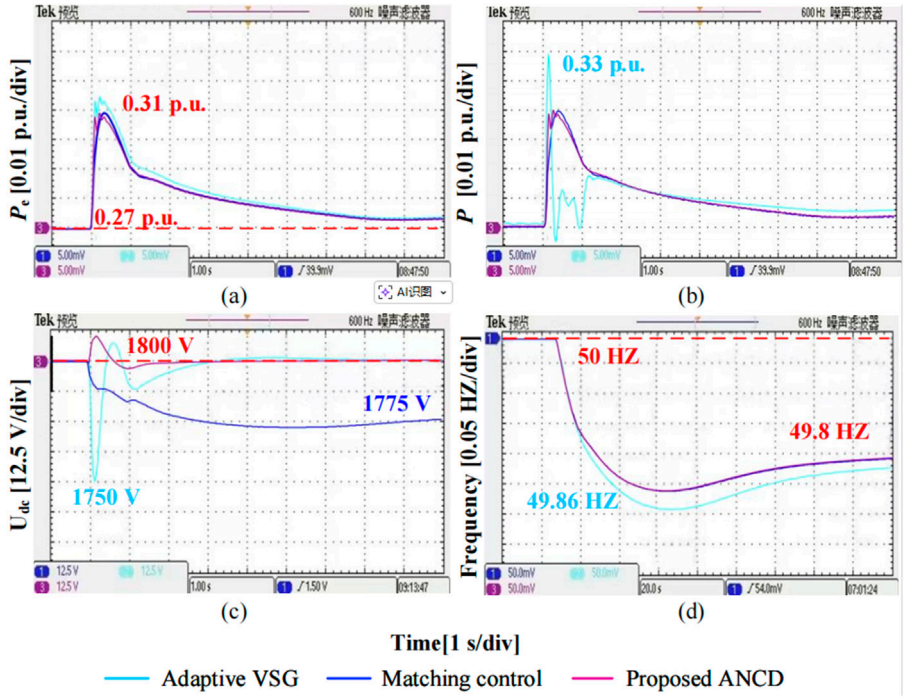


FIGURE 17
Simulation results of different grid-forming control strategies at a wind speed of 8 m/s. **(a)** WT electromagnetic power. **(b)** Grid-side converter power. **(c)** DC bus voltage. **(d)** System frequency.

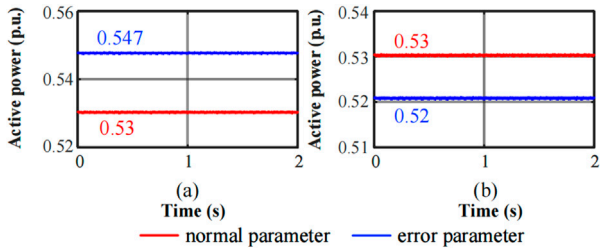


FIGURE 18
Variations in the active power output of the proposed strategy under parameter and environmental variations. **(a)** $\rho' = \rho + 0.075$ **(b)** $R' = R - 0.5$.

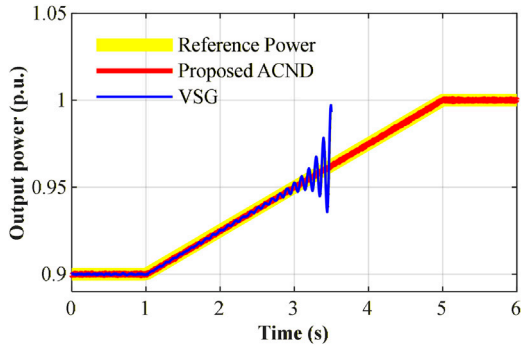


FIGURE 19
Experiment of adaptive parameter adjustment for ACND and VSG under different reserve power.

6.3 Analysis of the impacts of parameter variations and uncertainties

The method proposed in this paper relies on the calculation of the power coefficient, and thus is affected by the inherent parameters of the wind turbine, such as blade radius and air density. Meanwhile, signal fluctuations and time delays from the system will also exert an influence on the deloading accuracy of the wind turbine. This experiment aims to discuss the magnitude of the impacts of the above factors on the NDS. It can be seen from Equation 1 that the dynamic characteristics of the wind turbine are affected by air density and blade radius, and the variations of these two parameters with the environment and time will reduce the deloading accuracy. Under the conditions of a wind speed of 10 m/s and $k_{del} = 0.9$, when the air density increases from the standard atmospheric density of 1.225 kg/m^3 to 1.3 kg/m^3, the output power rises from 0.5302 to 0.547 with an error of 3.2%. By simulating blade wear to reduce the radius from 35 m to 34.5 m, the output power decreases from 0.5302 to 0.5202, corresponding to an error of −1.8%, as shown in Figure 18.

In this experiment, 5% noise and 5 ms time delay are introduced into the wind speed measurement sensor respectively, which induce output power errors of ±0.01% and ±0.015% accordingly. The influence of signal noise is negligible owing to the inertial characteristics of the pitch angle mechanical structure. Meanwhile, since the wind speed varies much slower than the signal time delay, the impact of the signal time delay can also be neglected.

6.4 Comparison of adaptive parameter adjustment for CND and VSG controllers

To verify the wide-range adjustability of the CND's inertia and droop coefficients, the initial power of the WT was set to 0.9 pu, and the turbine's output power gradually increased over time, meaning the reserve power of the WT gradually decreased. Both the CND controller and the VSG controller adaptively tuned their droop coefficient D_p and inertia time constant H according to Equations 20, 21.

As shown in Figure 19, during the gradual reduction of reserve power, the VSG controller's output power gradually began to oscillate and eventually became unstable. This is because the VSG's droop coefficient is coupled with the damping ratio—when the droop coefficient decreases with reserve power, the damping ratio also decreases, and oscillations occur when the damping ratio becomes too small. Meanwhile, the CND controller was able to follow the power commands while adjusting its inertia constant and droop coefficient. Even when the reserve power reached zero, meaning the CND controller's droop coefficient was zero, no oscillations occurred, verifying the wide-range adjustability and stability of the CND controller's parameters.

7 Conclusion

This paper addresses the issue of reserve power in WT and the balance between grid-forming controller output power and wind turbine power. A novel deloading strategy that coordinates rotor speed and pitch angle has been proposed, along with an adaptive CND controller suitable for deloaded wind turbines. The key contributions of this work are: 1) The novel deloading strategy enables power reservation and release by simply adjusting the deloading coefficient, eliminating the need for look-up tables to control rotor speed and pitch angle, and improving flexibility without requiring large data storage or model training. 2) The ACND controller's output power is well-matched with the wind turbine power, reducing DC bus voltage fluctuations. Compared to the VSG control algorithm, the ACND controller balances the input and output power of the wind turbine more effectively, and unlike matching control, the DC bus voltage is unaffected by system frequency. The proposed strategies have been verified through Matlab/Simulink simulations. Future research will focus on improving NDS to allow simultaneous RSC and PAC operation, enhancing the speed of reserve power release and reservation. Furthermore, we will continue to investigate the wind turbine power release characteristics and grid-forming algorithms to achieve better matching between the two.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author.

Author contributions

ZY: Supervision, Investigation, Conceptualization, Writing – review and editing. XK: Validation, Methodology, Writing – original draft, Investigation. XJ: Conceptualization, Validation, Writing – review and editing. QY: Software, Investigation, Writing – original draft. XS: Writing – original draft, Conceptualization.

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Conflict of interest

Authors ZY, XJ, and QY were employed by Huaneng International Power Jiangsu Energy Development Co., Ltd. Authors XK and XS were employed by State Grid Electric Power Research Institute/NARI Technology Co., Ltd.

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